
Question 1: Describe the Maxwell equations

Answer 1: $\nabla \times \bar{e} = -\frac{\partial \bar{b}}{\partial t}$

$$\nabla \cdot \bar{d} = \rho$$

$$\nabla \times \bar{h} = \bar{j} + \frac{\partial \bar{d}}{\partial t}$$

$$\nabla \cdot \bar{b} = 0$$

- $\bar{e}(\bar{r}, t)$ is ELECTRIC FIELD $[V/m]$
- $\bar{d}(\bar{r}, t)$ is the electric flux density $[C/m^2]$
- $\bar{h}(\bar{r}, t)$ is MAGNETIC FIELD $[A/m]$
- $\bar{b}(\bar{r}, t)$ is MAGNETIC FLUX DENSITY $[T = Wb/m^2]$
- $\rho(\bar{r}, t)$ is CHARGE DENSITY $[C/m^3]$
- $\bar{J}(\bar{r}, t)$ is ELECTRIC CURRENT DENSITY $[A/m^2]$
- $\bar{r} = (x, y, z)$

Assumptions: $\rho = 0, \bar{J} = 0$

Question 2: EM fields in harmonic domain

Answer 2: In many cases we can limit the study of EM fields to the harmonic signals. It is convenient to represent fields by their phaser using the following relation:

$$\bar{e}(\bar{r}, t) = \text{Re} [\bar{E}(\bar{r})e^{j\omega t}] \quad (1)$$

In this case we have no electric charges or current. Hence:

$$\nabla \times \bar{E} = -j\omega \bar{B} = -j\omega \mu \bar{H}$$

$$\nabla \cdot \bar{D} = 0$$

$$\nabla \times \bar{H} = j\omega \bar{D} = j\omega \epsilon \bar{E}$$

$$\nabla \cdot \bar{B} = 0$$

Question 3: Describe the constitutive relations

Answer 3: The relation between fields and flux densities depend on the medium where the fields propagate. In general we have:

$$\bar{D} = \epsilon \bar{E}$$

$$\bar{B} = \mu \bar{H}$$

where ϵ is the electric permittivity and μ is the magnetic permeability of the medium. $[F/m]$ and $[H/m]$ respectively.

Question 4: Describe Helmholtz equation

Answer 4: The Helmholtz equation is a second-order partial differential equation that describes the propagation of waves in a medium. Starting from the Maxwell equations in harmonic domain without charges and currents, we can derive the Helmholtz equation for the electric and magnetic fields.

Maxwell's Equations (Phasor Form)

$$\begin{aligned}\nabla \times \bar{E} &= -j\omega\mu\bar{H} & \nabla \cdot \bar{E} &= 0 \\ \nabla \times \bar{H} &= j\omega\varepsilon\bar{E} & \nabla \cdot \bar{H} &= 0\end{aligned}$$

Derivation of the Helmholtz Equation

1) Apply the curl to Faraday's Law:

$$\nabla \times (\nabla \times \bar{E}) = \nabla \times (-j\omega\mu\bar{H})$$

2) Due to the linearity of the operator, move the constants outside:

$$\nabla \times (\nabla \times \bar{E}) = -j\omega\mu(\nabla \times \bar{H})$$

3) Substitute the curl of $\bar{H}$ using the Ampère-Maxwell Law:

$$\nabla \times (\nabla \times \bar{E}) = -j\omega\mu(j\omega\varepsilon\bar{E})$$

4) Simplify the constant terms (given that $j^2 = -1$):

$$\nabla \times (\nabla \times \bar{E}) = \omega^2\mu\varepsilon\bar{E}$$

5) Use the vector identity $\nabla \times \nabla \times \bar{A} = \nabla(\nabla \cdot \bar{A}) - \nabla^2\bar{A}$:

$$\nabla(\nabla \cdot \bar{E}) - \nabla^2\bar{E} = \omega^2\mu\varepsilon\bar{E}$$

Since $\nabla \cdot \bar{E} = 0$ in a source-free medium, the gradient term vanishes:

$$-\nabla^2\bar{E} = \omega^2\mu\varepsilon\bar{E}$$

6) Rearrange the terms to obtain the Helmholtz Equation:

$$\nabla^2\bar{E} + \omega^2\mu\varepsilon\bar{E} = 0$$

For magnetic field the steps are the same.

Question 5: What is the advantage of using piecewise homogeneous media?

Answer 5: Piecewise homogeneous is a simplification that allows us to consider ϵ and μ as constants in a given region. But normally ϵ and μ are not constant, so we solve this problem by using boundary conditions.

Question 6: What are the boundary conditions?

Answer 6: We consider the thin surface between 2 different materials. We define the normal versor $\hat{n}$.

For flux densities

- **Density of magnetic flux:** normal component is always continuous

$$\hat{n} \cdot (\bar{B}_2 - \bar{B}_1) = 0 \quad (2)$$

- **Density of electric flux:** normal component is discontinuous

$$\hat{n} \cdot (\bar{D}_2 - \bar{D}_1) = \rho_s \quad (3)$$

For fields

- **Electric field:** tangential component is always continuous

$$\hat{n} \times (\bar{E}_2 - \bar{E}_1) = 0 \quad (4)$$

- **Magnetic field:** tangential component is discontinuous

$$\hat{n} \times (\bar{H}_2 - \bar{H}_1) = \bar{J}_s \quad (5)$$

For our assumptions, $\rho_s = 0$ and $\bar{J}_s = 0$.

Question 7: Describe the Poynting vector

Answer 7: In the time domain the Poynting vector is defined as:

$$\bar{P}(\bar{r}, t) = \bar{E}(\bar{r}, t) \times \bar{H}(\bar{r}, t) \quad (6)$$

where $\bar{E}$ is the electric field and $\bar{H}$ is the magnetic field.

While in the harmonic domain we have:

$$\bar{P}(\bar{r}) = \frac{1}{2} \bar{E}(\bar{r}) \times \bar{H}^*(\bar{r}) \quad (7)$$

where $\bar{H}^*$ is the complex conjugate of $\bar{H}$.

In other words, it represents the power flow per unit area.

Question 8: Describe the polarization of EM waves

Answer 8: A generic sinusoidal EM field can be written as:

$$\bar{e}(\bar{r}, t) = \sum_{n=1}^{3 \text{ dim}} C_n(\bar{r}) \cos(\omega t + \phi_n(\bar{r})) \hat{x}_n = \underbrace{c'(\bar{r}, t)}_{\bar{E}_{\text{Re}}} \cos(\omega t) - \underbrace{c''(\bar{r})}_{\bar{E}_{\text{Im}}} \sin(\omega t) \quad (8)$$

which clearly shows that the electric vector of a generic sinusoidal field lays at any time on a plane defined by the vectors $\bar{E}_{\text{Re}}$ and $\bar{E}_{\text{Im}}$.
